

2 The transport of respiratory gases involves blood plasma and red blood cells. Red blood cells contain the globular protein, haemoglobin.

(a) Describe features of a haemoglobin molecule that are typical of a globular protein, **other than** having an approximately spherical shape.

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..... [2]

(b) A number of substances are involved in the transport of respiratory gases.

Complete Table 2.1 by stating the name of the substance that matches the description of its role in the transport of respiratory gases.

The first row has been completed for you.

Table 2.1

substance	role in the transport of respiratory gases
carbon dioxide	In the capillaries of respiring tissues, this combines with water to form carbonic acid.
	In the capillaries of respiring tissues, this enters red blood cells through a membrane transport protein.
	In red blood cells, this is formed when carbon dioxide binds to haemoglobin.
	In alveolar capillaries, this combines with a hydrogen ion in red blood cells to form carbonic acid.
	In alveolar capillaries, this is formed in red blood cells as a result of oxygen binding to a haem group.
	In red blood cells, this combines with haemoglobin to form haemoglobinic acid.

[5]

[Total: 7]